

GWO Based Computation Capacity Optimization for Pinching Antennas Assisted MEC Systems

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Abstract—In this paper, a wireless powered mobile edge computing system enabled by pinching antennas is studied, and a algorithm for computation capacity optimization is proposed. By combining the grey wolf optimizer with successive convex approximation, the proposed algorithm jointly optimizes device transmit powers, time allocation, PAs' positions, and radiation control to maximize computation capacity. Simulation results show that the proposed PA-assisted design outperforms conventional antenna systems in computational performance.

Index Terms—Pinching antennas, wireless power transfer, mobile edge computing, grey wolf optimizer

I. INTRODUCTION

With the rapid development of the Internet of Things (IoT), many smart terminals are deployed in complex scenarios such as environmental monitoring, intelligent healthcare, and industrial production [1]. Due to strict size and cost constraints, most terminals are equipped with small batteries and limited processing capability, which makes reliable operation in computation-intensive applications difficult. Wireless-powered mobile edge computing, integrating wireless power transfer (WPT) with mobile edge computing (MEC) [2], has been proposed to address this issue: energy is wirelessly delivered to battery-limited devices, and intensive tasks are offloaded to nearby base station servers. However, system performance remains fundamentally constrained by wireless propagation loss, which degrades both energy transfer efficiency and computation capability.

In order to mitigate the aforementioned propagation loss, various flexible antenna technologies have been developed. Reconfigurable intelligent surfaces (RIS) [3]–[5] can create intelligent wireless environments and establish virtual line-of-sight (LoS) links by controlling the phase and amplitude of reflecting elements, but their performance is limited by the cascaded base-station (BS)–RIS link. Fluid antennas [6], [7] and movable antennas [8]–[10] improve link quality by changing antenna positions within several wavelengths and thus reducing spatial loss, but the small movement range restricts gains in large-scale non-line-of-sight (NLoS) scenarios.

To overcome these limitations, pinching antennas (PAs) were proposed as a highly reconfigurable antenna solution [11]. Based on the leaky-wave principle, PAs are formed by attaching tunable dielectric elements onto an extended dielectric waveguide, which enables controllable signal radiation at

arbitrary positions and flexible LoS construction [12], [13]. Benefiting from their high reconfigurability and low cost [14], PAs show strong potential for reconfigurable communications in MEC systems, but PAs channels involve both waveguide and free-space propagation, which leads to complicated analytical forms and highly non-convex optimization problems.

Motivated by these issues, this paper investigates a PA-assisted wireless-powered MEC system with NOMA, where terminals harvest downlink energy and offload uplink tasks via PAs along a dielectric waveguide [15], [16]. By adaptively adjusting PAs' positions, free-space path loss is reduced, improving both energy efficiency and communication capacity. We formulate a joint optimization of PAs' positions and communication parameters under position and energy constraints and solve this highly non-convex problem via alternating optimization: updates PAs' positions with the Grey Wolf Optimizer (GWO) [18] and utilize Successive Convex Approximation (SCA) [19] optimizes communication variables. As a swarm intelligence algorithm, GWO deploys search agents (i.e. wolves) in the function space and guides the remaining wolves to find the optimal solution in space through the three best positioned wolves. SCA is used to transform originally non-convex optimization problems into a series of easily solvable convex subproblems. Simulations demonstrate that the proposed design significantly improves computation capacity over conventional fixed-antenna systems.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As shown in Fig. 1, a PA-assisted wireless-powered MEC system is considered. The waveguide accommodates M PAs and serves K MEC devices. Each device is equipped with energy harvesting and storage modules, with the capability for both local computing and task offloading to the MEC server at the feeding point. To enable the pinching antennas (PAs) to be promptly available at the optimized locations, we adopt an engineering solution in which antennas are densely deployed along the waveguide, and only those PAs located at the algorithm-determined optimal positions remain in the clamped state, instead of requiring the PAs to move rapidly along the waveguide.

The system operates in time-division duplex (TDD). Each frame of duration T is divided into a downlink wireless power

transfer phase with duration τ_1 and an uplink task-offloading phase with duration τ_2 .

A. Network Geometry and Channel Model

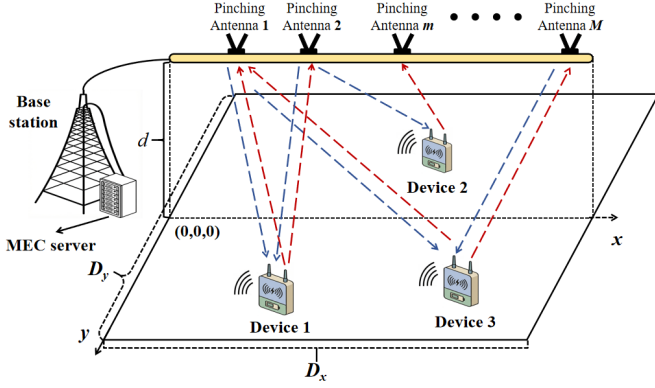


Fig. 1. Illustration of PA-assisted wireless-powered MEC systems.

Devices are randomly distributed in a rectangle of size $D_x \times D_y$. Taking the left edge as the y -axis and the horizontal midline as the x -axis, the position of device k is

$$\Psi_k = (x_k, y_k, 0), \quad k = 1, \dots, K. \quad (1)$$

The waveguide is deployed along the x -axis at height d , with the feeding point located at $\Psi_{P_0} = (0, 0, d)$, and total length D_x . The position of the m -th leaky-wave antenna is $\Psi_m^P = (x_m, 0, d)$, $m = 1, \dots, M$.

During the downlink WPT phase, the channel from the feeding point via the m -th PA to device k is modeled as [15]

$$h_{mk}^D = \underbrace{\frac{\eta \alpha_m}{\|\Psi_k - \Psi_m^P\|}}_{\text{free-space path loss}} \cdot \underbrace{e^{-j \frac{2\pi}{\lambda} \|\Psi_k - \Psi_m^P\|}}_{\text{free-space phase shift}} \cdot \underbrace{e^{-j \frac{2\pi}{\lambda_g} \|\Psi_0^P - \Psi_m^P\|}}_{\text{in-waveguide phase shift}}, \quad (2)$$

where Ψ_m^P is the position of the m -th PA in the downlink, $\eta = \frac{c}{4\pi f_c}$ denotes the reference channel gain, λ is the free-space wavelength, and $\lambda_g = \lambda/n_e$ is the guided wavelength with effective refractive index n_e . The coefficient α_m denotes the radiation factor of the m -th PA and characterizes the coupling strength between the dielectric particle and the waveguide. Due to energy conservation, the radiation factors satisfy $\sum_{m=1}^M \alpha_m^2 \leq 1$, and waveguide attenuation is assumed negligible [14].

Under the linear energy harvesting model [17], the harvested energy at device k in one frame is given by

$$E_k = P_B \beta \tau_1 \sum_{m=1}^M |h_{mk}^D|^2, \quad \forall k, \quad (3)$$

where P_B is the BS transmit power and $\beta \in (0, 1]$ is the energy conversion efficiency.

During the uplink phase, each device adopts task partitioning: part of the task is computed locally and the remaining

part is offloaded. The number of locally computed bits and the corresponding energy consumption at device k are

$$C_k^L = \frac{T f_k}{D}, \quad e_k = T \kappa f_k^3, \quad \forall k, \quad (4)$$

where f_k is the computation frequency (cycles/s), D is the computation intensity, and κ is the effective capacitance coefficient [17].

The uplink channel from device k to the feeding point via the m -th PA is expressed as

$$h_{mk}^U = \frac{\eta e^{-j \frac{2\pi}{\lambda} \|\Psi_k - \Psi_m^{PU}\|} \cdot e^{-j \frac{2\pi}{\lambda_g} \|\Psi_0^P - \Psi_m^{PU}\|}}{\|\Psi_k - \Psi_m^{PU}\|}, \quad (5)$$

where Ψ_m^{PU} is the position of the m -th PA in the uplink.

With NOMA, all devices share the same time-frequency resource. The total number of offloaded bits in one frame is given by

$$C^{\text{off}} = B \tau_2 \sum_{k=1}^K \log_2 \left(1 + \frac{p_k \sum_{m=1}^M |h_{mk}^U|^2}{\sum_{j=k+1}^K p_j \sum_{m=1}^M |h_{mj}^U|^2 + M \sigma_b^2} \right) \quad (6)$$

where B is the system bandwidth, p_k is the transmit power of device k , and σ_b^2 is the receiver noise power. The users are ordered such that $\sum_{m=1}^M |h_{m1}^U|^2 \geq \dots \geq \sum_{m=1}^M |h_{mK}^U|^2$ and SIC is applied at the receiver. Therefore, the total computation capacity of the system is

$$C_{\text{sum}} = C^{\text{off}} + \sum_{k=1}^K C_k^L. \quad (7)$$

B. Problem Formulation

By properly configuring the uplink and downlink PAs' positions Ψ_m^P , Ψ_m^{PU} and the radiation factors α_m , the path loss between the feeding point and the devices can be reduced. Further performance gains can be achieved by jointly optimizing device transmit powers p_k , local computation frequencies f_k , and the time allocation (τ_1, τ_2) .

The objective is to maximize the total computation capacity by jointly optimizing Ψ_m^P , Ψ_m^{PU} , α_m , p_k , f_k , τ_1 , and τ_2 :

$$\max_{\Psi_m^P, \Psi_m^{PU}, \alpha_m, p_k, \tau_1, \tau_2, f_k} C_{\text{sum}} \quad (8)$$

$$\text{s.t.} \quad |\Psi_m^P - \Psi_{m'}^P| \geq \Delta, \quad \forall m, m', m \neq m', \quad (8a)$$

$$|\Psi_m^{PU} - \Psi_{m'}^{PU}| \geq \Delta, \quad \forall m, m', m \neq m', \quad (8b)$$

$$x_m \in [0, D_x], \quad \forall m, \quad (8c)$$

$$p_k \tau_2 + e_k \leq E_k, \quad \forall k, \quad (8d)$$

$$\sum_{m=1}^M \alpha_m^2 \leq 1, \quad (8e)$$

$$\tau_1 + \tau_2 \leq T. \quad (8f)$$

where x_m^D and x_m^U denote the x -coordinates of the m -th PA in the downlink and uplink, respectively, and Δ is the minimum

spacing between adjacent PAs. The first three constraints guarantee feasible PAs deployment along the waveguide, the fourth constraint enforces the per-device energy budget, and the last two constraints ensure normalized radiation and feasible time allocation.

III. PROPOSED ALGORITHM

Since both the objective function C^{sum} and its constraints incorporate position variables of PAs, and there exists inherent coupling among Ψ_m^{PU} , p_k , and τ_1 , the joint optimization problem (8) exhibits non-convexity. The multitude of constraints complicates the optimization objective domain, significantly increasing computational complexity. To address this, we propose an alternating iterative optimization framework that optimizes different parameters step by step in one iteration, and use GWO and SCA to solve non-convex optimization problems.

A. Uplink and Downlink PAs Position Optimization

1) *Uplink PAs Position Optimization*: We first give other parameters in the optimization problem to optimize the uplink antenna position Ψ_m^{PU} . The optimization problem can be rephrased as:

$$\begin{aligned} \max_{\Psi_m^{PU}} \quad & \sum_{k=1}^K p_k \sum_{m=1}^M |h_{mk}^U|^2 \\ \text{s.t.} \quad & (8b), (8c). \end{aligned} \quad (9)$$

Since the search factor of GWO cannot actively comply with the constraints, we transform problem (8) into an unconstrained problem by adding a penalty term:

$$\min_{\Psi_m^{PU}} - \sum_{k=1}^K p_k \sum_{m=1}^M |h_{mk}^U|^2 + \mathbb{I}(\Psi_m^{PU}), \quad (10)$$

where $\mathbb{I}(\Psi_m^{PU})$ is the penalty term related to constraints (8b) and (8c), defined as follows

$$\mathbb{I}(\Psi_m^{PU}) = \begin{cases} 0, & \text{if (8b) and (8c) hold,} \\ +\infty, & \text{otherwise.} \end{cases} \quad (11)$$

In the proposed framework, each wolf represents a candidate antenna position vector $\Psi^U \triangleq [x_1^U, \dots, x_M^U]^T$, where x_m^U is the x -coordinate of the m -th leaky-wave antenna. At each GWO iteration, we maintain a population of N wolves. The current best three solutions with the smallest objective function value in (10) are denoted by Ψ_α^U , Ψ_β^U , Ψ_δ^U corresponding to the α , β , and δ wolves, respectively. The remaining wolves are treated as ω wolves.

The core idea of GWO is to iteratively move each wolf toward the three best wolves α , β , and δ using a set of adaptive coefficient vectors. Specifically, the relative position of a wolf with respect to a leader is modeled as

$$\mathbf{D}_\alpha = |\mathbf{C}_1 \odot \Psi_\alpha^U - \Psi_i^U|, \quad (12a)$$

$$\mathbf{D}_\beta = |\mathbf{C}_2 \odot \Psi_\beta^U - \Psi_i^U|, \quad (12b)$$

$$\mathbf{D}_\delta = |\mathbf{C}_3 \odot \Psi_\delta^U - \Psi_i^U|. \quad (12c)$$

where $\odot$ denotes element-wise multiplication, $|\cdot|$ is taken element-wise, and $\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3$ are coefficient vectors defined as $\mathbf{C}_l = 2\mathbf{r}_l$, $l \in \{1, 2, 3\}$, where $\mathbf{r}_l$ is random vector whose elements are independently drawn from the uniform distribution $\mathcal{U}(0, 1)$. The candidate positions relative to each leader are given by

$$\Psi_1^U = \Psi_\alpha^U - \mathbf{A}_1 \odot \mathbf{D}_\alpha, \quad (13a)$$

$$\Psi_2^U = \Psi_\beta^U - \mathbf{A}_2 \odot \mathbf{D}_\beta, \quad (13b)$$

$$\Psi_3^U = \Psi_\delta^U - \mathbf{A}_3 \odot \mathbf{D}_\delta, \quad (13c)$$

where the vectors $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are defined as $\mathbf{A}_l = 2\mathbf{a} \odot \mathbf{r}_l - \mathbf{a}$, $l \in \{1, 2, 3\}$, where $\mathbf{a}$ is a control vector whose elements linearly decrease from 2 to 0 during the optimization process. At the t -th iteration,

$$\mathbf{a}(t) = 2 \left(1 - \frac{t-1}{T_{\max}-1} \right) \mathbf{1}, \quad (14)$$

where $T_{\max}$ is the maximum number of GWO iterations and $\mathbf{1}$ is an all-ones vector. When $|\mathbf{A}_l| > 1$, the wolves explore the search space globally; when $|\mathbf{A}_l| < 1$, the wolves exploit the neighborhood of the current best solutions. The new position of the i -th wolf is finally updated as the average of the three candidate positions:

$$\Psi_i^U(t+1) = \frac{\Psi_1^U + \Psi_2^U + \Psi_3^U}{3}. \quad (15)$$

To strictly satisfy (8b)–(8c), we project the positions after each update:

- Boundary: clip x_m^U to $[0, D_x]$:

$$x_m^U \leftarrow \min\{\max\{x_m^U, 0\}, D_x\}, \quad \forall m. \quad (16)$$

- Spacing: if $\exists m \neq m'$ with $|x_m^U - x_{m'}^U| < \Delta$, slightly shift the two antennas along the waveguide so that $|x_m^U - x_{m'}^U| = \Delta$ while keeping them in $[0, D_x]$, and then re-evaluate the objective.

The combination of penalty-based fitness and projection confines the GWO search to the feasible set of (8b)–(8c) and yields physically realizable PAs layout. The GWO stops when the iteration index reaches $T_{\max}$ or the improvement of the best objective between two iterations is below ε . The final alpha wolf position Ψ_α^U is taken as the optimized uplink PAs placement and passed to the outer alternating-optimization framework.

2) *Downlink PAs Position Optimization*: For the downlink, the harvested energy E_k at each user is obtained from the downlink channel according to (2). Since each user achieves the best computation performance by fully using its harvested energy, constraint (8d) is tight and $p_k = \frac{1}{\tau_2}(E_k - e_k)$ holds at the optimum.

The downlink PAs position vector Ψ^{PD} is optimized by

$$\begin{aligned} \max_{\Psi^{PD}} \quad & \sum_{k=1}^K g_k \beta \tau_1 P_B \sum_{m=1}^M |h_{mk}^D|^2 \\ \text{s.t.} \quad & (8a), (8c), \end{aligned} \quad (17)$$

where $g_k = \left| \sum_{m=1}^M h_{mk}^U \right|^2$ denotes the uplink channel gain. This problem has the same structure as (8). Therefore, the above GWO-based procedure as in the uplink case is directly applied to obtain the optimized downlink PAs position.

B. Joint Optimization of Radiation Factors, Device Transmit Power, Time Allocation and Local Computational Frequency

In this subsection, the radiation factors $\{\alpha_m\}$ and device transmit powers $\{p_k\}$ are optimized. Define the radiation vector $\mathbf{w} = [\alpha_1, \dots, \alpha_M]^T$ and the channel vector between device k and all PAs as

$$\mathbf{u}_k = \left[\frac{e^{-j\frac{2\pi}{\lambda} \|\Psi_k - \Psi_1^{PU}\| - j\frac{2\pi}{\lambda_g} \|\Psi_0^P - \Psi_1^{PU}\|}}{\|\Psi_k - \Psi_1^{PU}\|}, \dots, \frac{e^{-j\frac{2\pi}{\lambda} \|\Psi_k - \Psi_M^{PU}\| - j\frac{2\pi}{\lambda_g} \|\Psi_0^P - \Psi_M^{PU}\|}}{\|\Psi_k - \Psi_M^{PU}\|} \right]^T. \quad (18)$$

Given $\tau_1, \tau_2, \{f_k\}, \Psi_M^{PU}, \Psi_M^{PD}$, and introducing auxiliary variables $\{t_k\}$, the subproblem is

$$\begin{aligned} \max_{\mathbf{w}, \{t_k\}} \quad & \sum_{k=1}^K g_k \beta_{\tau_1} P_B \eta^2 t_k \\ \text{s.t.} \quad & t_k \leq |\mathbf{u}_k^T \mathbf{w}|^2, \quad \forall k, \end{aligned} \quad (19)$$

(8e).

Problem (20) is non-convex. To solve this problem, we use SCA [19] to linearize the constraints around given point $\hat{\mathbf{w}}$ and obtain new expression of constraint conditions:

$$t_k \leq 2\hat{\mathbf{w}}^T \Re(\mathbf{u}_k^* \mathbf{u}_k^T) \mathbf{w} - \hat{\mathbf{w}}^T \Re(\mathbf{u}_k^* \mathbf{u}_k^T) \hat{\mathbf{w}}, \quad \forall k, \quad (20)$$

where $\Re(\cdot)$ denotes the real part operator. The optimization problem has convex constraints and can be solved by CVX solver. p_k is updated according to constraint (8d).

Then the downlink/uplink times τ_1, τ_2 and computing frequencies $\{f_k\}$ are optimized. Given $\Psi_M^{PU}, \Psi_M^{PD}, \{\alpha_m\}$ and $\{p_k\}$, the subproblem is

$$\begin{aligned} \max_{\tau_1, \tau_2, \{f_k\}} \quad & C^{\text{off}} + \sum_{k=1}^K C_k^L \\ \text{s.t.} \quad & (8d), (8f). \end{aligned} \quad (21)$$

Since it is convex, it is directly solved by CVX [20].

IV. SIMULATION RESULTS

In this section, the proposed system design and optimization approach are validated by simulations and compared with three baseline schemes: Conventional MIMO, Uniform PAs deployment and TDMA scheme.

The simulation area is set as $D_x \times D_y = 30 \text{ m} \times 10 \text{ m}$, and the waveguide height is $d = 3 \text{ m}$. The frame duration is $T = 1 \text{ s}$. The noise power spectral density is -174 dBm/Hz , and the bandwidth is $B = 100 \text{ MHz}$. The computation intensity is $D = 200 \text{ cycles/bit}$. Unless otherwise stated, the carrier frequency is $f_c = 29 \text{ GHz}$, the path-loss exponent is $n_e = 1.4$ [16], and $\kappa = 1 \times 10^{-28}$ [17].

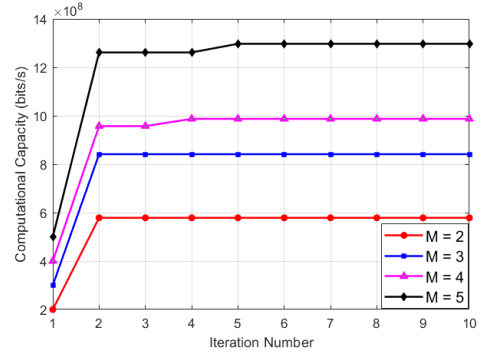


Fig. 2. Convergence performance of the proposed algorithm.

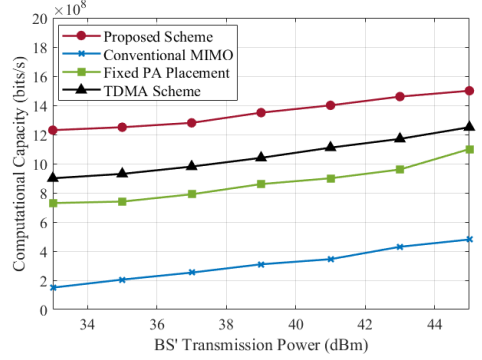


Fig. 3. Computation capacity versus BS transmit power P_B .

We first tested the convergence speed of the proposed algorithm under different PA quantities. As shown in Fig. 2, simulation results indicate that while the alternating optimization algorithm's convergence speed slows down with increasing PA numbers due to the expanded optimization problem dimensions, it still maintains relatively fast convergence speeds.

In Fig. 3, comparison of computation capacity versus BS transmit power P_B is given. Such figure illustrates that, due to reduced path loss via the waveguide, the PA-assisted schemes achieve higher computation capacity than conventional MIMO across all BS transmit powers. Moreover, optimizing PAs' positions and using NOMA further improves performance over uniform PAs placement and TDMA.

V. CONCLUSION

This paper proposes an innovative approach to integrate PAs into wireless-powered MEC systems, significantly enhancing computational performance. By employing tunable dielectric elements in extended waveguides to enable spatially reconfigurable signal transmission, the PA-assisted architecture effectively reduces large-scale path loss during downlink WPT and uplink task offloading phases. For the non-convex joint optimization problem involving time allocation, transmit power, and PAs positioning, we proposed an alternating algorithm combining GWO and SCA. Simulation results validate the effectiveness of the proposed design, demonstrating substantial computation performance improvements.

REFERENCES

- [1] D. C. Nguyen *et al.*, "6G Internet of Things: A Comprehensive Survey," *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 359–383, Jan. 2022.
- [2] F. Wang, J. Xu and S. Cui, "Optimal Energy Allocation and Task Offloading Policy for Wireless Powered Mobile Edge Computing Systems," *IEEE Transactions on Wireless Communications*, vol. 19, no. 4, pp. 2443–2459, Apr. 2020.
- [3] Q. Wu and R. Zhang, "Intelligent Reflecting Surface Enhanced Wireless Network via Joint Active and Passive Beamforming," *IEEE Transactions on Wireless Communications*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.
- [4] M. Hua, Q. Wu, W. Chen, Z. Fei, H. C. So and C. Yuen, "Intelligent Reflecting Surface-Assisted Localization: Performance Analysis and Algorithm Design," *IEEE Wireless Communications Letters*, vol. 13, no. 1, pp. 84–88, Jan. 2024.
- [5] G. Chen, Q. Wu, W. Chen, D. W. K. Ng and L. Hanzo, "IRS-Aided Wireless Powered MEC Systems: TDMA or NOMA for Computation Offloading?," *IEEE Transactions on Wireless Communications*, vol. 22, no. 2, pp. 1201–1218, Feb. 2023.
- [6] Y. Zhang and X. Chen, "Highly Pattern-Reconfigurable Antenna for Fluid Antenna System," 2025 IEEE International Symposium on Antennas and Propagation and North American Radio Science Meeting (AP-S/CNC-USNC-URSI), Ottawa, ON, Canada, 2025, pp. 2160–2163.
- [7] A. F. M. S. Shah, M. A. Karabulut, E. Cinar and K. Rabie, "Performance Analysis of Fluid Antenna Multiple Access (FAMA) for 6G," 2024 IEEE 15th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON), Berkeley, CA, USA, 2024, pp. 301–305.
- [8] L. Zhu, W. Ma and R. Zhang, "Movable Antennas for Wireless Communication: Opportunities and Challenges," *IEEE Communications Magazine*, vol. 62, no. 6, pp. 114–120, Jun. 2024.
- [9] L. Zhu, W. Ma and R. Zhang, "Modeling and Performance Analysis for Movable Antenna Enabled Wireless Communications," *IEEE Transactions on Wireless Communications*, vol. 23, no. 6, pp. 6234–6250, Jun. 2024.
- [10] Y. Gao, Q. Wu and W. Chen, "Joint Transmitter and Receiver Design for Movable Antenna Enhanced Multicast Communications," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, pp. 18186–18200, Dec. 2024.
- [11] H. O. Y. Suzuki and K. Kawai, "Pinching antenna: Using a dielectric waveguide as an antenna," *NTT DOCOMO Technical J.*, vol. 23, no. 3, pp. 5–12, Jan. 2022.
- [12] Z. Ding, R. Schober and H. Vincent Poor, "Flexible-Antenna Systems: A Pinching-Antenna Perspective," *IEEE Transactions on Communications*, vol. 73, no. 10, pp. 9236–9253, Oct. 2025.
- [13] C. Ouyang, Z. Wang, Y. Liu and Z. Ding, "Array Gain for Pinching-Antenna Systems (PASS)," *IEEE Communications Letters*, vol. 29, no. 6, pp. 1471–1475, Jun. 2025.
- [14] K. Wang, Z. Ding and R. Schober, "Antenna Activation for NOMA Assisted Pinching-Antenna Systems," *IEEE Wireless Communications Letters*, vol. 14, no. 5, pp. 1526–1530, Mar. 2025.
- [15] Y. Xu, Z. Ding and G. K. Karagiannidis, "Rate Maximization for Downlink Pinching-Antenna Systems," *IEEE Wireless Communications Letters*, vol. 14, no. 5, pp. 1431–1435, Mar. 2025.
- [16] S. A. Tegios, P. D. Diamantoulakis, Z. Ding and G. K. Karagiannidis, "Minimum Data Rate Maximization for Uplink Pinching-Antenna Systems," *IEEE Wireless Communications Letters*, vol. 14, no. 5, pp. 1516–1520, Mar. 2025.
- [17] B. Li, J. Liao, W. Wu and Y. Li, "Intelligent Reflecting Surface Assisted Secure Computation of Wireless Powered MEC System," *IEEE Transactions on Mobile Computing*, vol. 23, no. 4, pp. 3048–3059, Apr. 2024.
- [18] Z. Zhang, T. Xu, K. Zou, S. Tan and Z. Sun, "Multi-Objective Grey Wolf Optimizer Based on Improved Head Wolf Selection Strategy," 2024 43rd Chinese Control Conference (CCC), Kunming, China, 2024, pp. 1922–1927.
- [19] K. Ma, P. Liu, J. Li, C. Wang and Y. Yu, "Power Optimization Based on Successive Convex Approximation in Smart Grid Communications," 2018 5th International Conference on Information, Cybernetics, and Computational Social Systems (ICCSS), Hangzhou, China, 2018, pp. 478–482.
- [20] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.